PriceGuard-AI: A Decision Intelligence System for Price Monitoring and Prediction

Hackathon Project Documentation

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Abstract

Price volatility remains a critical challenge in developing urban environments, where supply chain disruptions, seasonal demand, and market inefficiencies lead to frequent and unpredictable fluctuations in the cost of essential goods. Consumers and small retailers often lack access to structured, predictive tools, forcing them to rely on guesswork and reactive purchasing behaviors. This paper presents the design, architecture, and implementation of PriceGuard-AI, a novel decision intelligence system engineered to transform raw, fragmented price data into actionable insights. Utilizing a robust Django backend, an interactive Streamlit dashboard, and a lightweight Artificial Intelligence (AI) module, the system tracks historical data, detects market trends, and generates human-readable purchasing recommendations (e.g., "Buy Now" or "Wait") accompanied by confidence intervals. Furthermore, the system addresses data scarcity through a validated, crowdsourced data ecosystem, supporting multilingual interfaces (English and Amharic) to ensure accessibility. By prioritizing explainability and data efficiency over complex, data-hungry models, PriceGuard-AI bridges the gap between passive data observation and active, informed decision-making in low-resource market environments.

1. Introduction

1.1 Background

Price volatility is a pervasive and destabilizing challenge in many real-world markets, particularly within rapidly developing urban environments such as Addis Ababa [1]. In these regions, the prices of essential goods and commodities can fluctuate dramatically on a daily or weekly basis. These variations are driven by a complex matrix of factors, including localized supply chain disruptions, fluctuating transportation costs, seasonal demand shifts, and inherent market inefficiencies [2]. Despite the severe economic impact of these fluctuations on daily life, the majority of consumers and small-scale retailers lack access to structured, analytical tools that could help them navigate and understand these dynamic changes.

Consequently, purchasing decisions in these environments are predominantly reactive rather than proactive. Users typically alter their buying habits only after a significant price increase has already occurred, relying heavily on anecdotal guesswork rather than empirical data [3]. PriceGuard-AI was conceptualized and developed to directly address this critical information gap. By transforming raw, unstructured price data into actionable decision intelligence, the system moves beyond mere data visualization. It actively analyzes historical pricing patterns, detects emerging trends, and provides users with clear, definitive recommendations—such as “Buy Now” or “Wait”—bolstered by transparent explanations and statistical confidence levels. To achieve this, the architecture integrates a robust backend API developed in Django, a highly interactive user-facing dashboard built with Streamlit, and a lightweight, specialized AI module dedicated to prediction and logical reasoning. Furthermore, the platform is designed with inclusivity in mind, featuring multilingual support (English and Amharic), comprehensive accessibility considerations, and rigorous data validation layers to manage user-contributed information.

1.2 Statement of the Problem

Consumers and small businesses operating in volatile markets face a multitude of interconnected challenges that hinder optimal price-based decision-making. The core issues can be delineated into several key areas:

- **Lack of structured price data:** Price information is highly fragmented, scattered across various physical locations and informal networks, and is rarely aggregated into a centralized, accessible repository [4].
- **No predictive insights:** While users may occasionally ascertain current market prices, they possess no systematic means to anticipate future price trajectories, leaving them vulnerable to sudden spikes.
- **High uncertainty in decision-making:** Without predictive context, buyers face a constant dilemma, unsure whether executing a purchase immediately or delaying it will yield better financial outcomes.
- **Limited transparency across locations:** Spatial arbitrage is common; the exact same product may command significantly different prices in adjacent neighborhoods due to localized monopolies or transportation bottlenecks [1].
- **Data scarcity and inconsistency:** High-quality, reliable, and continuous datasets are notoriously difficult to obtain for local, informal markets, rendering traditional big-data approaches ineffective.
- **Trust issues in user-generated data:** While crowdsourcing offers a solution to data scarcity, unvalidated user contributions are susceptible to inaccuracies, malicious manipulation, and noise [5].

PriceGuard-AI focuses its architecture and methodology on solving one overarching, critical problem: converting uncertain, sparse, and fragmented price information into clear, reliable, and actionable decisions for the end-user.

1.3 Objectives

1.3.1 General Objective

The primary objective of this project is to design, develop, and implement a data-efficient, AI-powered decision intelligence system that equips users with actionable price insights and robust decision support, specifically optimized for environments characterized by limited, noisy, and inconsistent data.

1.3.2 Specific Objectives

To realize the general objective, the project pursues the following specific goals:

- **System Development:** Build a comprehensive platform capable of tracking prices across diverse products, geographical locations, and temporal scales, while providing real-time trend analysis and short-term forecasting.
- **AI-Driven Decision Layer:** Implement an intelligent module that classifies market trends (e.g., increasing, decreasing, stable), recommends specific purchasing actions (Buy / Wait), and elucidates these decisions using human-readable, explainable reasoning.
- **User-Centric Interface Design:** Create an intuitive dashboard that facilitates price comparisons, supports multiple languages (specifically English and Amharic), and adheres to modern accessibility standards.
- **Robust Data Ecosystem:** Engineer a participatory data architecture that accepts user contributions, rigorously validates this data through statistical aggregation and anomaly detection, and iteratively improves model accuracy as user engagement scales.
- **User Engagement Mechanisms:** Introduce gamification and utility features, such as personalized watchlists, automated price alerts, and a contribution-based point system to incentivize continuous data entry.

2. Related Work / Literature Review

Price forecasting and market monitoring systems have been subjects of extensive study within the fields of econometrics and data science. Traditional approaches have heavily relied on time-series forecasting models. Techniques such as Autoregressive Integrated Moving Average (ARIMA) and Seasonal ARIMA (SARIMA) have been the gold standard for predicting macroeconomic trends and stock market fluctuations [6]. Concurrently, advanced machine learning models, including Random Forests, Support Vector Regression, and deep learning architectures like Long Short-Term Memory (LSTM) networks, have been deployed for complex regression and classification tasks in pricing [7]. Furthermore, commercial price comparison platforms have successfully aggregated data across digital vendors to provide consumers with real-time purchasing options.

However, when applied to developing urban markets and informal economies, these traditional systems exhibit severe limitations. Primarily, models like LSTMs and SARIMA require massive, continuous, and highly structured datasets to function effectively—a prerequisite that is fundamentally absent in environments like Addis Ababa [2]. Secondly, complex machine learning models often operate as "black boxes," lacking the interpretability required to build trust with everyday consumers [8]. Finally, commercial comparison platforms

are typically optimized for e-commerce and formal retail, failing to capture the localized, physical nature of informal markets.

Recent academic research has begun to pivot towards addressing these specific constraints. There is a growing emphasis on deploying lightweight, interpretable models that can maintain robust performance even on small or sparse datasets [4]. Additionally, the integration of Explainable AI (XAI) has been shown to significantly improve user trust and compliance with algorithmic recommendations [8]. To combat data scarcity, researchers are increasingly exploring crowdsourced data systems, utilizing mobile technology to gather real-time updates directly from consumers [5].

Positioning of PriceGuard-AI: PriceGuard-AI differentiates itself from existing literature and commercial tools by operating effectively under the constraint of limited data. Rather than outputting raw, probabilistic predictions that require user interpretation, it provides clear, actionable decisions. By integrating user-generated data with stringent validation mechanisms and prioritizing real-world usability and explainability over sheer model complexity, PriceGuard-AI represents a pragmatic evolution in consumer-facing economic tools.

3. Methodology

3.1 Datasets

Data Sources

The efficacy of PriceGuard-AI relies on a hybrid data acquisition strategy, utilizing two primary sources to overcome the "cold start" problem inherent in new platforms. Initially, the system is seeded with *Synthetic Demo Data*. This dataset was meticulously engineered to mimic real-world market behaviors observed in urban African markets, encompassing a variety of scenarios including sharp inflationary spikes, seasonal decreases, and stable pricing periods. The secondary, and ultimately primary, source is *User-Contributed Data*. Collected directly through the application's interface, this crowdsourced data is vital for capturing hyper-local, real-time price fluctuations and is used to continuously retrain and refine the system's prediction accuracy over time.

Data Structure

To ensure consistency and facilitate efficient querying within the Django backend, each data record adheres to a strict schema. A standard entry includes the *Product Name*, a hierarchical *Category* classification (e.g., Food, Electronics, Essentials), the recorded *Price* in local currency, the specific *Location* (geotagged or neighborhood-level), and a precise *Timestamp*. This structured approach allows for multidimensional analysis across time, space, and product type.

Data Validation

Given the reliance on crowdsourced information, maintaining data integrity is paramount. PriceGuard-AI employs a multi-layered validation protocol. First, *Aggregation* is utilized; by requiring multiple inputs for a specific product in a specific location before updating the public index, the system naturally dilutes the impact of erroneous entries [5]. Second, automated *Outlier Detection* algorithms (utilizing Interquartile Range and Z-score methodologies) flag extreme values that deviate significantly from historical norms. Third, a *User Confirmation* mechanism allows community members to upvote or verify recent entries. Finally, an *Admin Control* dashboard empowers moderators to manually review and purge suspicious or malicious data points.

Training, Validation, and Test Strategy

Adapting to the reality of limited data availability dictates a specialized approach to model evaluation. The strategy partitions the data chronologically rather than randomly. *Training* is conducted using historical records to establish baseline patterns. *Validation* utilizes recent trend data to tune the lightweight model parameters and thresholds. Finally, *Testing* is performed against the most recent, unseen entries to evaluate the model's real-world predictive capability and its responsiveness to sudden market shifts.

Preprocessing

Before data enters the AI module, it undergoes rigorous preprocessing. This includes *Data Cleaning* to remove null or invalid entries, and *Normalization* to standardize product names and categories (e.g., merging "Teft" and "Teff" into a single identifier). The data is then sorted chronologically. Crucially, the preprocessing pipeline includes *Feature Creation*, where raw prices are transformed into derived metrics such as price change percentages, categorical trend directions, and rolling volatility indicators, which serve as the primary inputs for the decision engine.

3.2 Techniques

Why AI in PriceGuard-AI?

The fundamental challenge addressed by this project is not merely the storage and retrieval of prices, but the comprehension of market dynamics and the generation of prescriptive advice. Without Artificial Intelligence, the system would be relegated to a passive data viewer—a digital ledger of historical facts. By integrating AI, the system actively detects latent trends, forecasts short-term movements, and synthesizes this information to recommend specific actions. Therefore, AI is the catalyst that transforms PriceGuard-AI from a standard database application into a proactive decision intelligence system.

Prediction Approach and Decision Logic

The prediction engine eschews data-heavy deep learning in favor of lightweight, statistical heuristics optimized for sparse datasets. It analyzes historical price patterns using weighted moving averages to calculate short-term trends and estimate the percentage change over upcoming temporal windows. The *Decision Logic* is deterministic and threshold-based, designed for maximum clarity: the system issues a "Buy Now"

recommendation if the predictive model indicates a statistically significant increasing trend (suggesting the user should purchase before prices rise further). Conversely, it issues a "Wait" recommendation if prices are stable or actively decreasing, allowing the user to potentially secure a better deal in the near future.

Explanation System and Confidence Estimation

To foster user trust, PriceGuard-AI incorporates an Explanation System based on Natural Language Generation (NLG) templates [8]. Instead of presenting a bare recommendation, each decision is accompanied by human-readable reasoning, such as “Prices increased 8% in the last 3 days” or “Stable trend observed across multiple locations.” Furthermore, the system calculates a *Confidence Estimation* (displayed as High, Medium, or Low). This metric is derived from data availability and variance; high volume, low-variance data yields high confidence, whereas sparse or conflicting data yields lower confidence.

Low Data Handling and Advanced Intelligence

A critical feature of the system is its graceful degradation in low-data scenarios. When insufficient data is detected for a specific item, the system avoids making misleading predictions, instead displaying warning messages. To mitigate this, it employs *Category-Based Modeling*, utilizing broader category-level trends (e.g., general inflation in "Food") to provide baseline insights when specific item data is sparse. Additionally, *Location Intelligence* algorithms compare prices across different neighborhoods, identifying spatial market inefficiencies and highlighting potential arbitrage opportunities for users.

4. Conclusion

PriceGuard-AI successfully demonstrates that highly impactful, user-centric AI systems do not inherently require massive datasets or computationally exhaustive models. By strategically focusing on clarity, real-world usability, and the specific constraints of developing urban markets, the system effectively transforms fragmented, uncertain price data into meaningful, actionable insights. The project delivers a robust platform that provides definitive purchasing recommendations, transparent and explainable reasoning, and a scalable, crowdsourced data ecosystem. Ultimately, PriceGuard-AI represents a vital paradigm shift from passive data observation to active, empowered decision-making, equipping consumers and small businesses with the intelligence necessary to navigate volatile and uncertain markets with confidence. Future enhancements, including the integration of external macroeconomic factors and real-time data pipelines, will further solidify its position as an indispensable economic tool.

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